

Readme:

Project Summary: Restaurant Menu System

What is this project?

It is a Java program to manage a restaurant menu. I built it to handle two things:

1. Single items (like a Burger).
2. Groups of items (like a Family Meal).

How does it work?

I used the Composite Design Pattern. This means I can treat a single item and a whole menu the same way.

The Classes I wrote:

- MenuComponent: The base interface for everything.
- MenuItem: For single items with a name and price.
- Menu: A "container" that holds other items. It calculates the Total Price by adding up everything inside it automatically.
- RestaurantSystemApp: Where I run the code and show the final menu.

Why I used this way?

- It's organized: Everything has its place.
- It's flexible: I can put a menu inside another menu easily.
- It's smart: The system calculates the price and handles the display (with spaces/indent) by itself.